

Maël TREMOUILLE

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Final-year engineering student at École polytechnique and ENSAE Paris, specializing in Data Science and Statistics. Seeking a PhD position in AI/Machine Learning starting Fall 2026. Strong experience in machine learning, statistical modeling and Python development.

Research interests: Mechanistic interpretability of large language and vision-language models; representation-level analysis of social-cognitive behaviour and of cross-modal binding.

EDUCATION

- École polytechnique & ENSAE Paris** — Double Degree 2024 – 2026
M.Sc. Data Science (X) & Engineering Degree in Data Science & Statistics (ENSAE)
Coursework: Deep Learning, Generative Models, Diffusion Models, Computer Vision, NLP, Advanced ML, Reinforcement Learning, Optimal Transport.
- ENSAI** — Engineering Degree, Statistics & Computer Science (top 5%) 2022 – 2024
Coursework: Supervised & Unsupervised Learning, Inferential Statistics, Measure Theory.
- Lycée Descartes** (Tours) — Preparatory Classes MPSI/MP 2020 – 2022

EXPERIENCE

- KU Leuven** (Belgium) — Research Intern May 2026 – Aug. 2026
- Research on representation-level steering of large language models (contrastive activation addition, inference-time intervention, INCLINE-style cross-lingual alignment) applied to cross-lingual and domain-specific tasks, within the INCLINE programme (Prof. Sien Moens).
 - Implementation, benchmarking, and ablation of these interventions on medical-NLP downstream tasks using decoder-only language models.
 - Goal: contribute to a research paper investigating when representation-level steering yields meaningful gains over vanilla decoding.
- ENSAE Junior Enterprise** — Data Science Consultant Feb. 2026 – Apr. 2026
- Designed a multi-agent system for automated operational risk assessment.
 - Pipeline orchestrating LLM agents (GPT-4o, o3) via LangGraph for document analysis and decision-making.
 - Delivered a production-ready Python package with configurable architecture.
- Groupe BPCE** — Research Data Science Intern Jun. 2025 – Sep. 2025
- Reviewed and implemented research papers on Positive-Unlabeled (PU) learning.
 - Designed and delivered a Python package for internal scoring and classification.
 - Presented results to the Group Scientific Committee (100+ attendees).

PROJECTS & CONTRIBUTIONS

- Study of the CANDI Model (Continuous And Discrete Diffusion)** Feb. 2026
- Analyzed Temporal Dissonance theory for noise scheduling in diffusion models.
 - Adapted the implementation for training on WikiText-2 with NVIDIA A100 GPU ($|V| \approx 30\,000$).
 - Evaluated the crystallization mechanism for non-autoregressive text generation.
- Adversarial Robustness and Model Capacity** Oct. 2025 – Jan. 2026
- Re-implemented min-max optimization to analyze architectural capacity (WideResNet vs CNN).
 - Benchmarked on CIFAR-10/GTSRB under iterative L_∞ (PGD, MIM) and L_0 attacks.
 - Optimized the accuracy-robustness trade-off using stochastic training as a regularizer.
- Surrogate Modeling Toolbox (SMT)** — Open-source Contributor 2024
- Implemented Gaussian process variance re-interpolation methods for Bayesian optimization.
 - Contributed optimization modules to the open-source Python library.
 - Unit testing and documentation to ensure library robustness.

SKILLS

Programming: Python, R, SQL, Git, Java, Docker, $\text{\LaTeX}$.

Machine Learning: Deep Learning (PyTorch, TF), GNN, Time Series, Boosting (CatBoost, XGB), scikit-learn.

LANGUAGES & INTERESTS

Languages: French (native), English (C1 – TOEIC 955).

Sports: 10 years of wrestling/BJJ (former assistant coach), Boxing.

Reading: Absurdist philosophy (Camus), Russian novels (Dostoevsky). Writing reviews and personal essays.